

SOURABH AMBARSHETTI

Senior Full Stack Engineer · Forward Deployed AI Engineer

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 7+ years of production experience architecting high-throughput backend services, distributed data pipelines, and multi-tenant SaaS platforms for global enterprise clients including **Lenovo, HP, Dell, and Hitachi Astemo**. Proven track record diagnosing deep system bottlenecks—slashing enterprise quote ingestion latency by **77%** and eliminating 100% of V8 heap memory crashes on large datasets. Skilled in defensive programming, enterprise integrations, and zero-defect delivery across MEAN/Full-Stack ecosystems, currently expanding into **Forward Deployed Engineering (FDE) and Applied AI agent architectures**.

TECHNICAL SKILLS MATRIX

Backend & Distributed Systems: Node.js (v18/v20), TypeScript, Express.js, NestJS, LoopBack 3, REST APIs, Webhook Architecture, Asynchronous Worker Pools, Stream Processing, V8 Heap Memory Profiling, Microservices Architecture.

Databases & Data Engineering: PostgreSQL, MS SQL Server, MySQL, MongoDB, Redis, Query Tuning, Index Optimization, ACID Transactions, Row-Level Locking, ExcelJS Bulk Ingestion ETL, Relational Schema Modeling.

Applied AI & LLM Systems: Python, FastAPI, Retrieval-Augmented Generation (RAG), Vector Databases (pgvector, Qdrant), Dense Embeddings, Deterministic Tool Calling, Pydantic Schema Validation, Semantic Chunking, BM25 Hybrid Search, Ragas Evaluation.

Frontend & Mobile Engineering: Angular (v12, v14, v18), RxJS Reactive Streams, TypeScript, Ionic 6 (Cross-Platform Mobile), Angular Material, React, Next.js, Tailwind CSS, Responsive Web Design.

System Design & Resilience: Defensive Programming, Circuit Breaker Patterns, Exponential Backoff & Retries, Idempotent API Design, Zero Data Corruption Guarantees, Role-Based Access Control (RBAC).

DevOps, Cloud & Tooling: AWS (EC2, S3, RDS), Docker Containerization, CI/CD Pipelines, Git Version Control, Linux, Nginx, Agile/Scrum, Enterprise QA/UAT Staging Delivery.

PROFESSIONAL EXPERIENCE

Senior Full Stack Developer — PyxTech Pvt. Ltd.

Oct 2024 — Present | Pune, India

- Architect and scale enterprise SaaS platforms powering FMV-driven pricing (**PyxFMV**), trading (**PyxTrade**), and vendor bidding (**PyxQuote**) for the US refurbished electronics market, serving Fortune 500 clients including **Lenovo, HP, and Dell**.
- Re-engineered the **Bulk Quote Ingestion Engine** from monolithic in-memory parsing to stream-based chunking with bounded worker concurrency, dropping execution times on 1,200-row enterprise spreadsheets from **~15 minutes to 3.4 minutes (-77% latency reduction)**.
- Diagnosed an unconstrained Excel worksheet bounds issue that generated **3,496 unexpected phantom empty chunks**, engineering defensive spatial coordinate validation that permanently eliminated server heap memory crashes.
- Implemented end-to-end pricing, bidding, shipping logistics, and MRB dispute handling workflows with transactional batch commits and strict QA/UAT SLAs.
- Collaborated directly with enterprise client stakeholders, mentoring developers and establishing automated regression test suites.
- Spearheading the architectural integration of Applied AI workflows (RAG, agent tool dispatching) into core enterprise backend services.

PROFESSIONAL EXPERIENCE (CONTINUED)

Senior Full Stack Developer (MEAN Stack) — CodeNgin Technologies Pvt. Ltd. Sep 2023 — Oct 2024 | Pune, India

- Led full-stack architecture and backend RESTful microservices for high-concurrency enterprise web applications using Angular, Node.js, Express.js, LoopBack 3, MongoDB, and MySQL.
- Implemented complex business rule validation engines and secure third-party webhook integrations with automated exponential retry mechanisms and cryptographic payload verification.
- Established standardized technical documentation and architectural design patterns, accelerating engineer onboarding and knowledge transfer across engineering pods.
- Applied clean architecture principles and design patterns to streamline feature releases, conducting thorough code reviews and mentoring junior developers on defensive coding practices.

Senior Software Engineer (Promoted from Intern) — Primus Techsystems Pvt. Ltd. Oct 2019 — Sep 2023 (4.4 Yrs) | Pune, India

- Engineered the Arthsetu Ventures Loan Origination System (LOS) from scratch with automated Aadhaar/PAN KYC, Credit Bureau scoring APIs, and a custom Loan Eligibility Rule Engine; cut loan approval turnaround from 48+ hours to sub-15 minutes and deployed cross-platform mobile apps to Google Play Store & Apple App Store.
- Architected the Hitachi Astemo Dealer Management System (DMS) and companion Product Catalogue mobile applications, synchronizing multi-branch inventory and sales analytics across regional dealership networks and reducing stock allocation discrepancies by 92%.
- Built an end-to-end Construction Management ERP for SCON Projects from scratch, featuring site status telemetry, contractor KYC verification, milestone progress tracking, and automated budget variance monitoring.
- Designed normalized relational database schemas (MySQL, MS SQL) with indexing strategies sustaining sub-100ms query performance under peak production loads.
- Progressed from Intern Software Developer (Jun 2019 – Aug 2019) to Senior Software Engineer through consistent technical ownership, production support excellence, and zero-defect enterprise deliveries.

FEATURED ENTERPRISE PROJECTS & APPLIED AI ARCHITECTURES

High-Throughput Bulk Quote Engine & Memory Optimization PyxTech / Lenovo, HP, Dell

Asynchronous stream-processing pipeline handling 1,200+ row vendor quote files with worker thread concurrency, defensive spatial coordinate bounds pruning, and transactional batch writes. Capped memory consumption under 250MB, eliminated out-of-memory crashes, and reduced processing latency by 77%.

Enterprise Knowledge Assistant (Production RAG Architecture) Applied AI / FDE Blueprint

Production-grade Retrieval-Augmented Generation system with semantic document chunking, hybrid retrieval (pgvector dense vector embeddings + BM25 sparse lexical search), citation attribution, and RAG Triad hallucination evaluation metrics (Context Relevance, Groundedness, Answer Relevance). Built with Python, FastAPI, and LangChain.

Deterministic Autonomous Support & Action Agent Applied AI / Agent Architecture

Controlled agentic workflow with structured tool-dispatch protocols, Pydantic schema validation, sandboxed execution, prompt injection defenses, human-in-the-loop approval gates for destructive mutations, and comprehensive audit telemetry. Built with Python and LangGraph.

EDUCATION & ACADEMIC CREDENTIALS

Master of Computer Application (MCA) — Pune, India Graduated 2020

Advanced computing curriculum covering distributed software architectures, relational database management systems, advanced algorithms, and enterprise web applications.

Bachelor of Computer Application (BCA) — Solapur, India Graduated 2017

Foundational computer science studies in object-oriented programming, data structures, software engineering life cycles, operating systems, and full-stack web technologies.